PGSL PDF Toolkit

Indian Institute of Technology Ropar

1. Executive Summary & Overview

The PGSL PDF Toolkit is a lightweight, responsive desktop application developed for comprehensive PDF document management. Designed to meet academic and professional standards at IIT Ropar, the system combines high-performance PDF manipulation with an intuitive three-panel graphical user interface.

Key capabilities include non-destructive page rotation, reordering, deletion, insertion, merging, watermark application, plain text extraction, and optical character recognition (OCR).

2. Core Requirements

- FR-01: Document preview with responsive thumbnail strip and navigation.
- FR-02: Page-level manipulation (reorder, rotate, delete, insert).
- FR-03: Multi-document batch processing (merge PDFs, image to PDF).
- FR-04: Advanced tools (watermark overlay, compression, plain text export).
- FR-05: Integrated OCR for scanned documents across Indic and English scripts.

Architecture & Technical Design

The toolkit architecture strictly decouples graphical user interface components from core document processing engines. The system is composed of four principal subsystems:

1. Explorer Panel (Left): Directory navigation, file tree representation, and multi-file selection for batch operations.

2. Preview Panel (Center): High-fidelity rendering via pypdfium2, smooth zooming, and thumbnail-based page selection with multi-select support.

3. Toolkit Panel (Right): Vertically stacked control groups providing accessible action buttons, numerical spinboxes, and parameter sliders.

4. Session Hub & Worker: Central signal-driven communication coordinator ensuring panels never directly depend on one another. Long-running operations execute asynchronously in dedicated worker threads, providing non-blocking responsiveness and cancel support.

Testing, Verification & Quality Standards

All modifications undergo automated verification using pytest and ruff. Tests are run in headless offscreen environments (QT_QPA_PLATFORM=offscreen) to enable seamless continuous integration on server instances.

Key quality standards:

- Complete preservation of original document files (immutable working copy model).
- Undo stack supporting history traversal of the last 10 applied operations.
- Progress reporting with user cancellation checkpoints at every processing cycle.
- Robust exception handling for encrypted, corrupted, or unsupported media formats.